PART 6

Vehicles and pivots

What a “vehicle” is in data terms. It is not a list of N identical pivots but a few distinct **roles**, together with how the body gets its position and angle, and how a rig plugs into the mechanism.

6.1 Vehicle families
6.2 Pivot roles
6.3 Where the root goes
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6.1 Vehicle families

Family	Track contact	Mechanism needed
Mine cart / tram with one truck	One short rigid wheelbase	Root only (or two axle points if the wheelbase is visible on curves)
Four-wheel wagon	Two axles fixed to the frame	Two **axle points**; the frame aims between them
Bogie coach / subway car	Two swivelling bogies	Two **tracked pivots**; the body rides on both
Steam locomotive (e.g. 2-4-2)	Rigid driving wheels + swinging pony/trailing trucks	Driver **axle points** set the frame; **radial** trucks swing about pins on the frame
Tender (two bogies)	Two bogies	Same as a bogie coach, coupled to the loco by a drawbar
Articulated (Mallet, Garratt, articulated tram)	Several frames hinged together	Several bodies, each a sub-vehicle. A consist with zero gap and a hinge

6.2 Pivot roles

The v8 pivot list treats every pivot the same (“offset in metres, own Follow Path”). That is correct for bogies but not enough for locomotives. The vehicle-architecture session already identified three roles, and adding a fourth covers everything:

One vehicle can mix all four roles

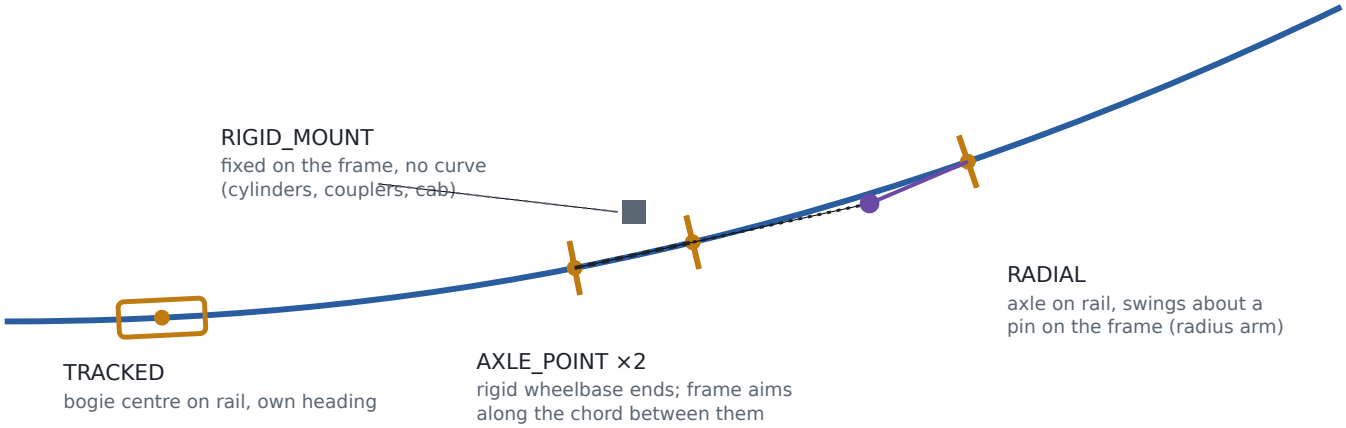


Figure 6.1 Pivot roles. Only TRACKED, AXLE_POINT and RADIAL sample the rail. RIGID_MOUNT is just a bone on the frame.

Role	Mechanism object	How its transform is found
TRACKED	Empty + Follow Path (as today)	Route point at *d + offset*, with that point's heading and tilt.
AXLE_POINT	Empty + Follow Path	Same as TRACKED. Only its **position** is used; the frame's heading comes from the pair.
RADIAL	Empty + Follow Path (axle) + stored `pin_offset`	Axle on rail; the truck bone sits at the axle and **Damped Tracks** toward the pin on the frame.
RIGID_MOUNT	None (bone in the rig)	Child of the frame bone.

6.3 Where the root goes

Option A: root at the vehicle's centre

The root sits midway between the two defining points (bogie centres, or the ends of the rigid wheelbase). Pivot offsets are symmetric (±). It looks natural in the viewport and matches how most models are built. Coupler positions are stored as `front_coupler` / `rear_coupler` distances in the definition.

Option B: root at the front coupler

Consist maths becomes one running number, but every pivot offset is negative and the gizmo sits at the nose. It is awkward to author, and flipping the vehicle's facing moves the reference point.

RECOMMENDATION

Option A (centre), plus coupler distances in the definition. The consist code computes spacing as `rear_coupler(car n) + gap + front_coupler(car n+1)`, and the user only enters the gap (often 0 when couplers touch). That keeps the “front-referenced chain” idea from session 4 without making the vehicle awkward to author.

6.4 Body orientation and the chord effect

A long car on a curve does **not** hug the rail: its body is a straight line between its two bogies, so its middle sits **inside** the curve and its ends overhang **outside**. Today's root rides the rail itself, so a mesh parented straight to the root clips through the inside of tight curves.

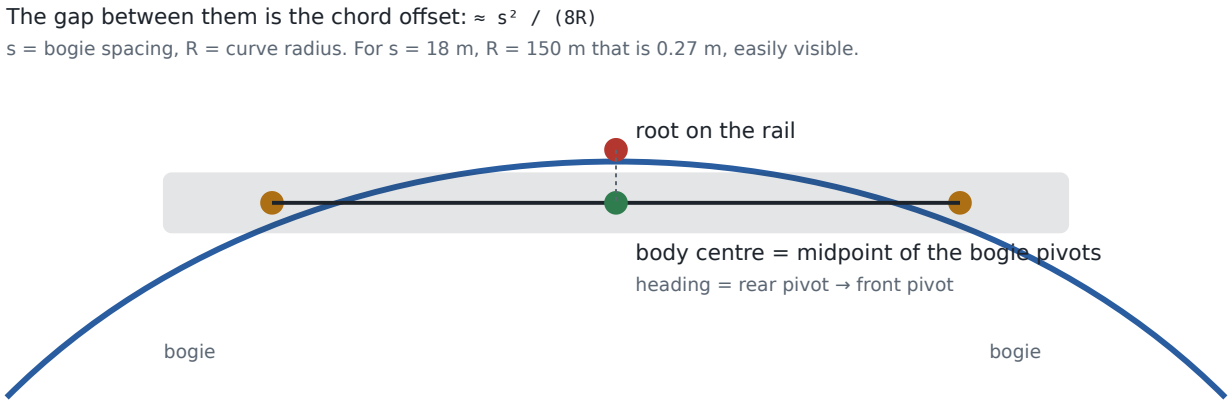


Figure 6.2 The body should come from the **pivots**, not the root. This is also how real rail vehicles behave.

Built with native constraints on the body bone

```
BODY bone
  Copy Location   → Pivot_Front Empty           influence 1.0
  Copy Location   → Pivot_Rear  Empty           influence 0.5   (= midpoint of the two)
  Damped Track    → Pivot_Front Empty  (Y)      heading along the chord
  Copy Rotation   → Pivot_Front (roll only, mix ADD, influence 0.5)
  Copy Rotation   → Pivot_Rear  (roll only, mix ADD, influence 0.5)   averaged banking
```

For a steam loco the same pattern uses the two **driver AXLE_POINTs** instead of bogies (Part 10).

6.5 Pivots crossing curves

The known gap (v8 §3.3): pivots are placed on the root's curve and clamp at its ends. Two ways to close it:

Approach	How	When
Short term: reuse the wagon resolver	Call `_track_it_resolve_route_position(root, root_distance + pivot.offset)` for each pivot, exactly as `_track_it_sync_consist_wagons` already does for wagons, then set that curve's constraint to influence 1 on the pivot Empty.	Now, small change
Target: the occupied path	Pivots are points on the train's occupied path (Part 5). Nothing special is needed.	Phase 3

6.6 Binding a rig to the pivots

The Pivot Data panel already stores `rig_object` and `rig_bone` (inert). A **Bind Rig** operator turns that into constraints on the pose bones, and **Unbind** removes exactly those constraints (tagged by name prefix `TI_`):

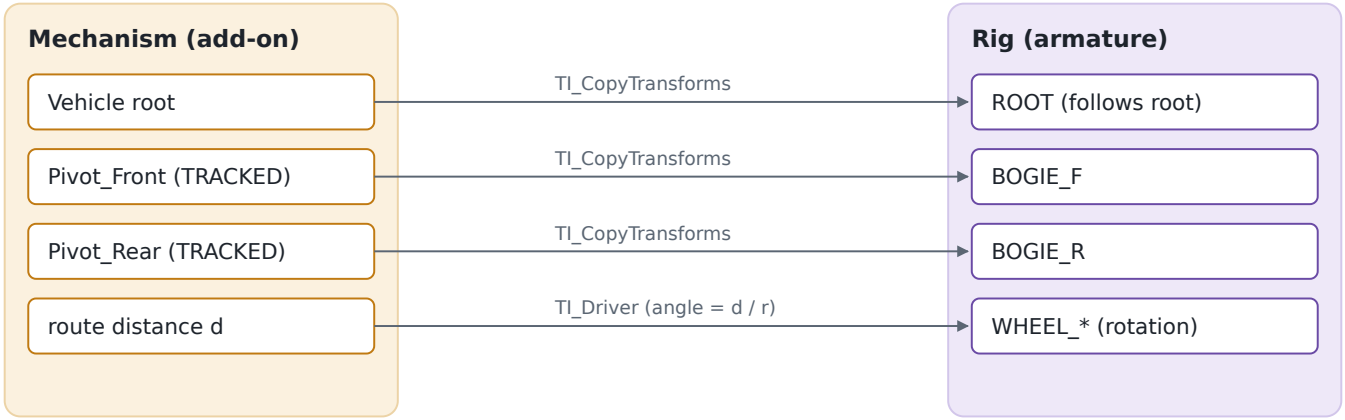


Figure 6.3 Binding is one-way (mechanism → rig) and fully removable. The artist's own constraints and keys on the rig are never touched.

BONE VS OBJECT

Pivot Empties should be **hidden and unselectable** in finished assets (the earlier plan), so nobody grabs them by accident. The Pivot Data panel keeps a “Show mechanism” eye toggle for debugging.

6.7 Facing and running backwards

A locomotive can run tender-first, and a coach can be turned end for end. Add a per-vehicle **facing** (+1 / −1):

- Pivot offsets and coupler ends are multiplied by facing when placing points.
- Follow Path on the root and pivots uses `forward_axis = 'FORWARD_Y'` when facing = +1 and `'TRACK_NEGATIVE_Y'` when −1, so the model turns 180° with no rig change.
- Wheel rotation sign flips automatically, because it is computed from distance × facing.
- This is the “direction flipping system” saved for later in session 5. It is a small addition once pivots are role-based.

6.8 The Vehicle Definition

Everything above that describes the *model* (not its current position) belongs in a definition saved with the asset:

```
VehicleDef "LNER_242T_Proto"
  type: LOCOMOTIVE          length_over_couplers: 11.80 m
  front_coupler: +5.90 m    rear_coupler: -5.90 m      facing_default: +1
  pivots:
    LEAD_TRUCK   role RADIAL      offset +3.95  pin_offset +2.60  bone TRUCK_F
    DRIVER_1     role AXLE_POINT  offset +1.05                    bone AXLE_D1
    DRIVER_2     role AXLE_POINT  offset -1.05                    bone AXLE_D2
    TRAIL_TRUCK  role RADIAL      offset -3.95  pin_offset -2.60  bone TRUCK_R
  frame: aims DRIVER_2 → DRIVER_1, bone FRAME
  wheel_groups:
    drivers   bones [WHL_D1_L, WHL_D1_R, WHL_D2_L, WHL_D2_R]  diameter 1.73 m  mode AUTO
    trucks    bones [WHL_F_L, WHL_F_R, WHL_T_L, WHL_T_R]      diameter 0.97 m  mode AUTO
  secondary: sway 0.4, hunting 0.2, bounce 0.3, joint_spacing 18.3 m
```

Numbers are illustrative (Part 10 works through a real layout). Stored as a JSON string property on the asset's root object, with a matching PropertyGroup for editing in the UI.